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Pune Vidyarthi Griha's
COLLEGE OF ENGINEERING, TECHNOLOGY AND MANAGEMENT, PUNE-09

Excellence in Engineering Education

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Unit Wise Question Bank

Department: Computer **Semester: I** **Academic Year: 2025 - 2026**

Class:SE **Div:** **Date:13/09/2025**

Course: Data Structure

Unit No. - name	Q. No.	Question	Marks	CO	Blooms Level
2 Searching and sorting	Q.1	Explain Linear search and binary search with example. State its time complexity and compare linear and binary search(time and space complexity)	6	2	
	Q.2	Write an algorithm for searching an element using binary search. Discuss the time complexity of algorithm in best case and worst case.	6	2	
	Q.3	. Write an algorithm for Fibonacci search and find out its time complexity.	6	2	
	.Q.4	Explain insertion sort algorithm and sort the given list using insertion sort : 1) List : 7, 4, 10, 6, 3, 12, 1, 8, 2, 15, 9, 5 2) List : 55, 85, 45, 11, 34, 05, 89, 99, 67	6	2	
	.Q.5	Write pseudo Python code for Bubble sort	6	2	
	Q.6	Write a pseudo code for binary search apply your algorithm on the following no.to stored in array to search no. 23 and 100 . 9,17,23,40,45,52,58,80,85,95,00	6	2	
	Q.7	Sort a given array using bubble sort method:[64,34,25,12,22,11,90] Show the step by step execution for all the passes highlighting the swap and No swap situation .how many passes are required to sort an array of the elements using bubble	6	2	

Study material provided by: V. B. L

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		sort?			
	Q.8	Enlist different searching methods.write short note on i) Fibonacci search Index sequential search.	6	2	
	Q.9	Explain in brief the different searching techniques. What is the time complexity of each of them.	6	2	
	Q.10	Sort the following numbers step by step using insertion sort 55, 85, 45, 11, 34, 5, 89, 99, 67 Comment on time complexity of Insertion sort.	6	2	
	Q.11	Explain in brief any three searching techniques. What is the time complexity of these techniques?	6	2	
	Q.12	Write an algorithm to search an element in array using binary search technique. Show stepwise search of Key 10 in 1, 2, 3, 4, 5, 6, 7, 8, 9, 10,11. Is there any pre-processing of data required before starting binary search or what is its precursor requirement?	6	2	
	Q.13	Sort given array by using selection sort method 50, 23, 03, 18, 9, 01, 70, 21, 20, 6, 40, 04. Show step by step execution of all passes. What is the best and worst case time complexity of	6	2	

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		selection sort?			

Course Teacher

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